



Inspiring Excellence

EEE 433L

Power Systems II Laboratory

Semester Project

Spring 2026

Full Marks- 50

Problem Statement: You will have to solve the load-flow problem for a typical 13-bus power system which is shown in Figure 1, based on the data provided for this power system in page-3 of this pdf.

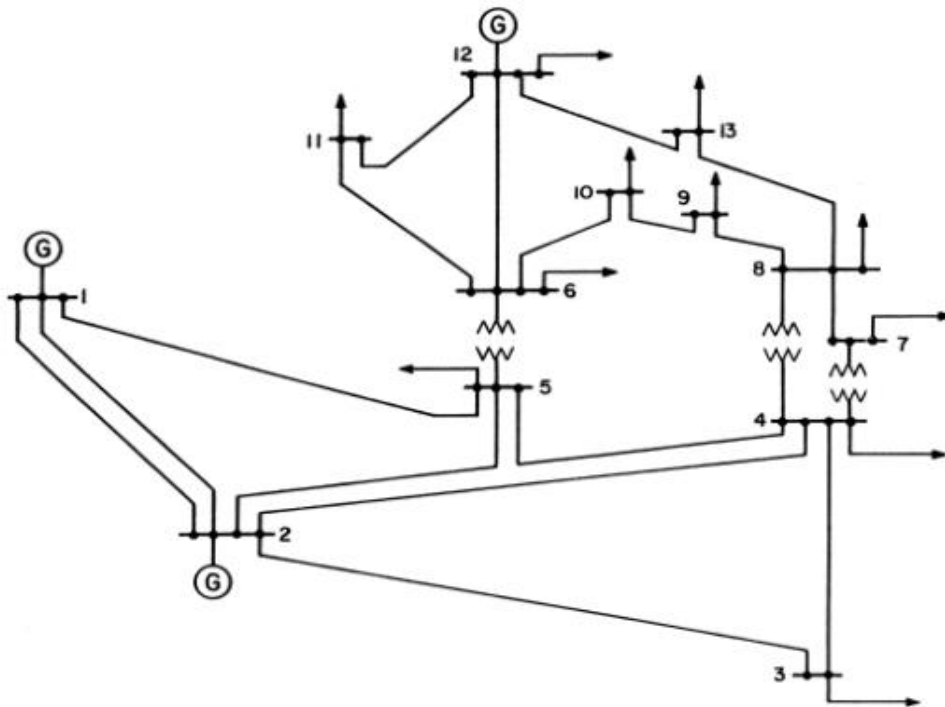


Figure 1. A single-line diagram of 13 Bus Power System

Answer the following questions:

- A. Conduct a load flow study on the power system shown in Figure 1 with the help of MATLAB code using **Gauss-Seidel** and **Newton-Raphson** methods. Which technique does give a better load flow solution in your opinion? Use **relevant mathematical figures of merits** to justify your answer. You must prepare a table with all the relevant parameters (Voltage, Power, line flows, line losses etc.) of the 13 bus system. A sample table has been shown.

Bus No.	Bus Type	Gauss Seidel				Newton Raphson			
		P	Q	V	δ	P	Q	V	δ

- B. A solar PV source is available which is capable of supplying **A0** MW load to the system. You have to connect this solar PV source in such a way that the cumulative line-loss in the system is minimum. **Which bus satisfies this requirement?** *You cannot connect the solar PV to your slack-bus.* Prepare a table with the relevant information to justify your answer. A sample is shown below:

Solar PV Bus	Total loss Before connection	Total loss after connection
2		
3		
4		

Note: “A” refers to your group number.

- C. By constructing the given 13 bus system in PV simulator, run the simulation for the cases as mentioned in Question A and B. Analyze the results and write your observations.

Deliverables:

- A Comprehensive Project Report with a cover-page which must include-
 - Some relevant theory
 - Answers to the given questions
 - Relevant Calculations
 - Work Distribution among group members
 - A concluding remark mentioning difficulties, pitfalls, potential solutions to problems encountered etc.
 - *Before submission*, make sure that the cover page has the signature and consent of all group members.
- Prepare the Excel File (as shown in class) for the bus-system given in this project. You must extract data from this excel file to use in your projects.
- All the MATLAB codes used to solve the given problems. Codes must be properly indented. Add *comments* in your code for ease of understanding. *Successful implementation of the code must be demonstrated on project submission day.*
- The simulation files of the PV simulator. *Successful implementation of the code must be demonstrated on project submission day.*
- Attach a Google Drive link in your project report containing source codes and all user defined functions. Name those code files in such a way that it is clearly understandable to which file is for what purpose.

AI Policy:

- Use of AI is heavily discouraged and might result in penalties.

13-Bus System Data

System Base: 100 MVA

(1) Bus Data

Bus No.	Nominal Voltage (kV)	Type	Real Power Load (MW)	Reactive Power Load (MVAR)
1	138	Slack Bus	--	--
2	138	PV	--	--
3	138	PQ	95	20
4	138	PQ	45	0
5	138	PQ	10	2
6	66	PQ	11	7.5
7	66	PQ	$K \times L^1$	$---$ ²
8	66	PQ	25	7
9	66	PQ	20	6
10	66	PQ	20	5
11	66	PQ	30	10
12	66	None	13.5	6
13	66	PQ	20	5

¹ L = Last digit of any of the group member's Student ID and K = Second last digit of your ID. If any is 0, use 1.

² Load power factor at bus 7 should be maintained at 0.9 lagging.

(2) Generator Data

Bus No.	P _g	V _g (pu)	Q _{max} (MVAR)	Q _{min} (MVAR)
1	Slack (not specified)	1.06	-	-
2	100	1.02	50	-40
12	20	1.01	100	-60

(3) Branch Data

From	To	R (pu)	X (pu)	B (pu)
Bus 1	Bus 2	0.01938	0.05917	0.05280
Bus 1	Bus 5	0.05403	0.22304	0.04920
Bus 2	Bus 3	0.04699	0.19797	0.04380
Bus 2	Bus 4	0.05811	0.17632	0.03400
Bus 2	Bus 5	0.05695	0.17388	0.03460
Bus 3	Bus 4	0.06701	0.17103	0.01280
Bus 4	Bus 5	0.01335	0.04211	0
Bus 6	Bus 10	0.09498	0.19890	0
Bus 6	Bus 11	0.12291	0.25581	0
Bus 6	Bus 12	0.06615	0.13027	0
Bus 7	Bus 8	0	0.11001	0
Bus 8	Bus 9	0.03181	0.08450	0
Bus 8	Bus 13	0.12711	0.27038	0
Bus 9	Bus 10	0.08205	0.19207	0
Bus 11	Bus 12	0.22092	0.19988	0
Bus 12	Bus 13	0.17093	0.34802	0

(4) Transformer Data

From	To	R (pu)	X (pu)	B (pu)
Bus 4	Bus 7	0	0.20912	0
Bus 4	Bus 8	0	0.55618	0
Bus 5	Bus 6	0	0.25202	0

Rubrics for this Project:

Sl.	Performance Criteria	Assessment Breakdown			
		Unsatisfactory	Developing	Satisfactory	Exemplary
1	Organization and extraction of bus-data from excel files	1	3	4	5
2	Perform Load Flow analysis using Gauss-seidel and Newton Raphson	4	6	8	10
3	Analyze the advantage and disadvantages from different load-flow solvers	4	6	8	10
4	Optimize a design by minimizing the line-loss using Solar PV system	1	3	4	5
5	Perform Load Flow analysis using PV Simulator and compare the result with load flow results from Matlab	4	6	8	10
6	Explain the difficulties and associated solutions encountered in load-flow problems.	4	6	8	10